

VIRAL DISEASE OF CANINE

		1. RABIES (Hydrophobia, Lyssa, Lytta or Rage)	2. Canine Distemper (Hard Pad Disease)
Definition		• Zoonotic, acute, highly fatal, progressive viral disease. • Caused by RNA virus (Rhabdoviridae family of the genus Lyssavirus). • Affects carnivores and bats, (any mammal can be affected). • Ch. by nervous signs, paralysis, coma & death.	• Highly contagious poly-systemic viral disease of dogs. • Ch. By diphasic fever, leukopenia, skin hyperkeratosis, GIT & respiratory tract and neurological complication. • 25-75% of susceptible dogs become sub clinically infected (long lasting immunity).
Aetiology		<i>Rabies virus, Lyssavirus & Rhabdoviridae.</i> • Large non-segmented, -ve RNA “bullet”. • Fragile: Susceptible to common disinfectant. • Dies in dried saliva in few hours. • Inactivated by boiling, direct sun light. • Persist in infected brain tissue for 7-10. d at room temperature and for several weeks at 4°C. • Virus is present (intermittent) in saliva of carnivora for 5.d before appearance of signs. Virus has 3 strains (1). Street strain (Isolated natural cases H or A). (2). Fixed strain (Attenuated lab. strain). (3). Flurry strain (human’s strains which is serially passage in different host system).	• CDV genus Morbillivirus" PPR and cattle plague" family Paramyxo. • Large ssRNA virus (one serotype). • Sensitive to UV radiation, heat, detergents and lipid solvents. • Survive for several days at temp. below zero and at -65°C for at least 7 years. Predisposing factors • Immunosuppression, poor feeding, debility, Vit A deficiency, parasitism and air draughts.
Epidemiology	Distribution:	Worldwide and endemic in Egypt.	
	Host rang:	• (All mammals, including domestic and non-domestic animals and humans). • Young more than adult. • Less common guinea pigs, rabbits, and pig • Opossums are very resistant. • Wildlife, raccoons, skunks, mongoose, bats are major reservoirs (immune carriers) inapparent infection.	• “Wide range Carnivora" dogs, foxes, raccoon, ferret, wolves, mink and skunk. • 3-6.m age are more susceptible (weaning & loss of maternal immunity). • Cats and pigs may be infected (bronchopneumonia). • Reported in non-human primates with high mortality rates (potential zoonotic risk of CDV in humans).
	Seasonal incidence	Late summer and autumn because of large scale movement of wildlife at mating and pursuit of food.	
	Transmission:	Source: infected saliva of rabid animal. Mode: Primary, biting of rabid animals. Occasionally, open wound or mucous membrane exposure to saliva or CNS tissue of rabid animal.	Source: Body tissues and secretions “respiratory (abundant) and conjunctival exudate, saliva, feces and urine for up to 2-3 m. post infection”. Mode: • Primary, inhalation. • Contact with contaminated objects. • Mechanically by flies and insects • Transplacental or in utero infection of puppies
Economic and zoonotic impact:		• Rabies considered a very dangerous disease with significant economic losses due to deaths' in animals and human. • The disease is zoonotic. N.B: Epidemiologically urban rabies refer to rabies of domestic a and pets while sylvatic rabies where wildlife are involved	Loss of dog's function and deaths of valuable dogs.
Pathogenesis		It is still not fully understood. 1. After biting of dog by infected animal or wound contamination by the virus, spreading occur through the body via the bloodstream. 2. Then the virus travels to the spinal cord and brain, at which point clinical signs of rabies often appear in the infected animal. 3. Virus spreads from the brain to the salivary glands.	1. After infection by inhalation, the virus multiplies in tissue macrophages. 2. Spreading within 1 d via the lymphatics to the tonsils and respiratory lymph nodes, resulting in severe immunosuppression. 3. Within 2-4 d., other lymphoid tissues become infected. 4. By day 6, the gastrointestinal mucosa, hepatic Kupffer cells and spleen are infected. 5. Further spread occurs by cell-associated viraemia to other epithelial cells and the CNS. 6. Viral virulence, host age and immunity play important role in the outcome of infection: a. Strong immunity, the virus fails to infect epith. tissues and viremia cease with elimination of the virus within 14.d and complete recovery occurs. b. Weak immunity, rapid dissemination of the virus occurs to epithelium of most organs as respiratory, GIT, eye, and CNS.

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Clinical Signs		I.P: From 2 weeks to 6 months depending on (Distance between site of bite and CNS, severity of bite, virulence & source and quantity of virus in saliva. Course: 1-10 d. Morbidity Low rate. Mortality High rate.	I.P: 2-9 d. Course: 10 d.- several months Morbidity Variable. Mortality Variable.		
Clinical Signs	Disease stages/ Forms	A. Prodromal 1. Subtle temperament changes & mild fever 2. Self-mutilation at bite site with slow blink reflexes B. Furious 1. Increasingly restless and irritable and attack inanimate objects. 2. Visual and auditory stimuli may trigger episodes of aggression, vocalizing with continues barking. 3. Walk aimlessly and depraved appetite. 4. Later loss of muscular coordination & generalized seizures. C. Paralytic 1. Dog is quiet and not irritable and do not bite until provoked, progressive ascending paralysis of throat and masseter. 2. Laryngeal/pharyngeal paralysis leads to drooling, and difficulty eating, drinking and breathing (hydrophobia). 3. The paralysis progress rapidly to all parts of the body. 4. Finally, coma and respiratory paralysis then death within hours.	A. Acute Systemic Form 1. Occurs 2–3 weeks post-infection. 2. Diphasic fever, depression, anorexia, mucopurulent oculo-nasal disch., cough, dyspnoea, vomiting and diarrhoea (may be bloody). 3. The virus is found in every secretion and excretion of the body. B. Chronic Nervous Disease 1. Concurrent or follow systemic disease within 2–3 weeks. 2. Abnormal behavior, convulsions or seizures, blindness, paresis or paralysis, incoordination, and circling. 3. "Chewing gum fits" type convulsive seizures cha. by chewing movements of the jaw with salivation occurs in dogs developing polioencephalomalacia. 4. Most animals die 2–4 w. after infection. 5. Hyperkeratosis of nose and foot pads with pustular dermatitis of lower abdomen (dandruff throughout the coat) is common in dogs suffering from neurological disease. 6. Chorea myoclonus: force involuntary neuromuscular twitching (jerking) of the muscles as in the legs or facial muscles, (specific). 7. It occurs due to local irritation of lower motor neurons of spinal cord or cranial nerve. 8. Can be present while dogs is walk or commonly while sleeping with involuntary defecation and urination. 9. It can be present in absence of other neurological signs.	C. Neonatal Infection Form 1. Occurs with or without neurological signs. 2. Infection of puppies before eruption of permanent dentition: severe damage of their enamel. 3. Dental enema become irregular in appearance or there is enamel hypoplasia. D. Transplacental Infection Form 1. Bitch may show inapparent infection, abortion, or birth of weak puppies 2. infected fetus may develop CNS signs during first 4-6 w. 3. Permanent immunodeficiencies occurs in survival puppies. E. Ocular form 1. The virus in the optic nerve and retina. 2. Optic neuritis: Ch. by blindness with dilated unresponsive pupils. 3. Dege. & Necrosis of retina produce gray to pink irregular densities on the eye, with atrophy and complete retinal detachment. 4. Circumscribed hyper-reflective areas termed " gold medallion " lesions which are characteristic of previous CDV infection. F. Complications Secondary viral, bacterial and or parasitic infections of the skin, digestive and respiratory tract "Immunosuppression".	
		P/M lesion		There are no pathognomonic lesions except congestion and edema of meninges and brain. Foreign materials in stomach of rabid cattle indicating depraved appetite	
		Field Diagnosis		Depends on case history, clinical signs, and P/M lesions.	
		Lab. Diagnosis	A. Sample: Head as a whole or brain, and smear or slices from brain and spinal cord. Salivary glands, saliva and nasal discharges. B. Laboratory Procedures: 1. Virus isolation on cell culture (CPE after 18- 24.h) "nuclei from the periphery to the center". 2. Microscopic examination of tissue smears after staining with seller's stain to detect intracytoplasmic inclusion bodies or Negri bodies (red or purplish). 3. Immunohistochemistry and IF to detect the virus in brain tissue. Serum 4. Mice inoculation: I/C injection of suckling mice with 10 % of brain tissue suspension (incubation period takes 11-12. d) paralysis. 5. Cerebrospinal fluid examination: it may have ↑ in protein and nucleated cell. 6. Serological tests as Virus Neutralization Test.	A. Sample (on ice or formalin): 1. Transtracheal or pharyngeal washing. 2. Smears or scrapings from conjunctiva & hard Pad. 3. Urine, CSF, tonsils, skin, uveal tissues. 4. CNS, spleen, lymph nodes, stomach, lung, duodenum, bladder, respiratory and genital epithelium. 5. Blood & serum. B. Laboratory Procedures: 1. Virus Isolation on cell culture (CPE after 2-5 d) "Giant cell formation". 2. Molecular Assays: Using (RT) PCR assay, nested PCR and real-time PCR, (highly sensitive and specific).	3. Serological Assays: (IFAT), ELISA and SNT. 4. Serum Biochemical Analysis: Decrease in albumin and increased in alpha and gamma globulin in adult. 5. Marked Hyperglobulinemia in puppies. 6. Haematology: lymphopenia, thrombocytopenia, regenerative anemia, 7. Histopathology: Acidophilic I.C & I.N I.B in the affected cells. 8. CSF Analysis: Increased in protein and cell count especially lymphocytes and anti-CDV antibody (IgG or IgM). 9. Animal Inoculation: I/C injection in mice, ferrets and hamster producing CNS signs. 10. Radiology: secondary bronchopneumonia.

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Differential Diagnosis (DDx)	All causes of Encephalitis or neurological diseases as 1. Canine distemper. 2. Infectious canine hepatitis. 3. Trauma. 4. Brain abscess or tumors. Also, the disease during Prodromal Phase is confused with 1. Obstruction in the throat. 2. Foreign body lodged between teeth. 3. Ingestion of irritant substances.	All causes of encephalitis or neurological diseases as 1. Rabies. 2. Infectious canine hepatitis. 3. Trauma. 4. Brain abscess or tumors. Also, 1. Leptospirosis. 2. Lead poisoning. 3. Toxoplasmosis. 4. Bacterial gastroenteritis. 5. Ehrlichiosis. 6. Coccidiosis.
Treatment	• No treatment should be attempted after appearance of signs. • Post exposure prophylactic measures: Immediately after exposure or biting, irrigation of the wound by 20 % soaps and water may prevent virus attachment, suture of deep wounds with infiltration with anti-rabies immune sera. • Post exposure vacc.: Useless (anti-rabies only)	A. Hygienic treatment: 1. Infected animal kept in clean warm and free of drafts. 2. Oculonasal discharges should be removed. 3. Food and water should be discontinued if vomiting and diarrhea is present. 4. Cleaning and disinfection of dog kennel. 5. Dead animal should be hygienically disposed. B. Medicated treatment: (No specific treatment). 1. Broad spectrum antibiotic as ampicillin, and synulox. 2. Ringer's solution given I/V or S/C. 3. Vit. B to replace those lost and to stimulate appetite. 4. I/V ascorbic acid, immune sera (10-30 ml), 5. Antiemetic, antipyretic, antinflammatory, and antidiarrheal drugs. 6. Analgesic and anticonvulsants after onset of systemic disease and prior to development of neurologic seizures. 7. Glucocorticoid (cortisol) may have variable success in controlling blindness and pupillary dilation. 8. Prognosis of the disease is generally bad.
Control	1. The main goal of animal's rabies control is to reduce or prevent infection of humans. 2. Destruction or elimination of stray dogs & cats and other wildlife population: 3. Imported dogs and cats, should vaccinated on arrival and put in quarantine. 4. Cautions should be taken in handling of rabid animals and infected materials. 5. Management of human bitten by rapid animals. 6. Vaccination of humans under high risk as veterinarian or farm workers (pre-exposure vaccination) with immediate using of immune sera or vaccination.	Segregation of infected dogs and treat them symptomatically and destruction all source of infection.
Vaccination	• Living attenuated egg adapted vaccines (flurry low egg passage) , used in puppies of age more than 3.m, I/M, immunity reach to two years, • Inactivated cell cultures vaccine (ERA viral strain), I/M, give immunity for 6-12. m., the above vaccines are produced and used in Egypt. • Wildlife can be vaccinated orally by baits vaccine.	• Living attenuated or inactivated vaccines singly or in combination with other canine vaccines. • Two doses with 3-4 weeks intervals, giving immunity 6 m -1 years. • Puppies from non-vaccinated bitch are vaccinate for first time at 1-4 w. age and at 6-16 w. age from vaccinated dam.

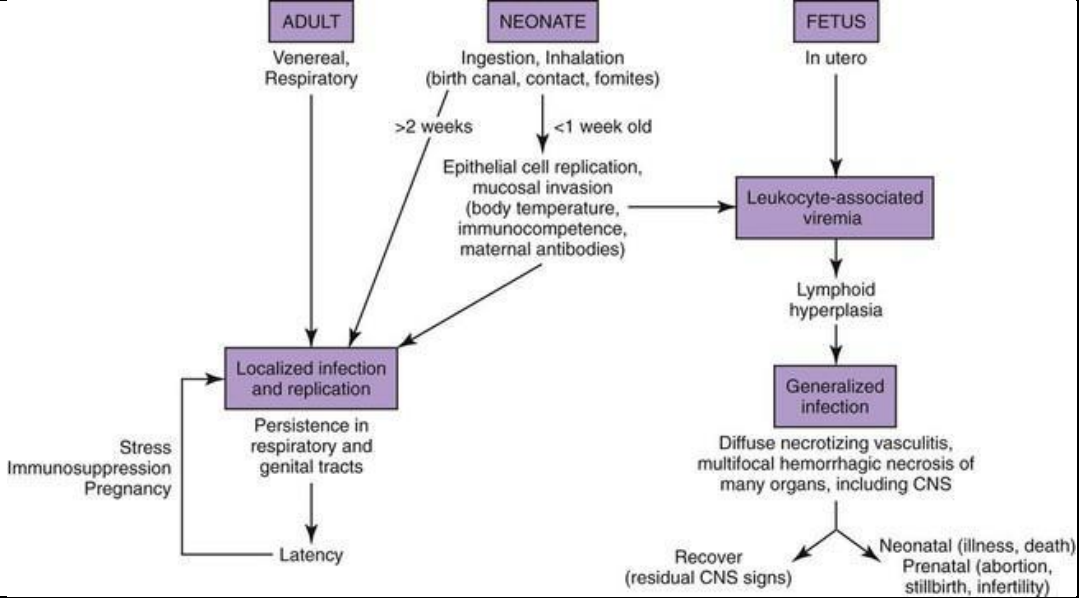
		3. Canine Parvovirus infection (CPV)	4. Canine coronavirus enteritis (CCVE)
Definition		Contagious infectious viral disease of dogs ch. by two different forms: Intestinal Form (more common stomach and intestines), which is characterized by anorexia, vomiting, diarrhea and weight loss. Cardiac Form (less common) attacks the heart muscles of very young puppies, often leading to death.	- Highly contagious intestinal disease (upper two thirds of the small intestine and local lymph nodes) of dogs. - Usually mild (enteritis and diarrhea) except after complication with others viruses (parvo) become more serious. - Some deaths in non-vaccinated puppies.
Aetiology		- CPV-2, family Parvoviridae. - Non-enveloped ssDNA virus. - Resistant to many common detergents and disinfectants. - Persist indoors at room temperature for a few weeks; outdoors for many months, if protected from sunlight and desiccation. Predisposing factors - Stress (eg, from weaning, overcrowding, malnutrition, etc), - Concurrent intestinal parasitism, or enteric pathogen infection (eg, <i>Clostridium</i> spp, <i>Campylobacter</i> spp, <i>Salmonella</i> spp, <i>Giardia</i> spp, <i>Coronavirus</i>) have been associated with more severe clinical illness.	- Canine coronavirus (CCV), RNA (remain in the body and shed into the feces for up to six months). - Closely related to the Feline Enteric Coronavirus (FIP), an intestinal virus that affects cats. 2 serotypes (CCV1) poorly grow in culture & ill-defined receptors & (CCV2) readily in culture and use the APN receptor. - Inactivated by most detergent and lipid solvent. Predisposing factors - Stress caused by over-intensive training, overcrowding and generally unsanitary conditions ↑ a dog's susceptibility to a CCV infection.
Epidemiology	Distribution:	Worldwide and reported in Egypt	Worldwide and reported in Egypt
	Host rang:	(Dogs, foxes, wolves and coyotes) - Puppies (6 weeks to 6 months) more susceptible. - Puppies less 6 w (inutero) of age take cardiac form while more than 6 w take intestinal form. - Rottweilers, Doberman, and German Shepherd dogs have been described to be at increased risk.	- Dogs all ages. - More in housed groups in a kennel.
	Seasonal incidence	None.	None
	Transmission:	A) Sources: Body secretions of dogs during acute stages of the disease as saliva & feces. B) Mode of transmission: - Ingestion. - Inhalation (rare). - Inutero infection.	Source: feces of infected dogs. Mode: Ingestion of contaminated food and water.
Economic and zoonotic impact:		Loss of dog's function and deaths of valuable dogs.	Loss of dog's function and deaths of valuable dogs. Zoonotic impact.
Pathogenesis		- Virus is shed in the feces of infected dogs within 4–5 days of exposure (often before clinical signs develop), throughout the period of illness, and for ~10 days after clinical recovery. - Infection is followed by replication in lymphoid tissue of the oropharynx then hematogenous dissemination. - Lymphopenia and neutropenia develop secondary to destruction lymphopoietic tissues. Intestinal form: (more than 6 w of age) Destruction of the intestinal crypt epithelium results in epithelial necrosis, villous atrophy, impaired absorptive capacity. Cardiac form (less than 6 w of age) myocardial infection, necrosis, and myocarditis presenting as acute cardiopulmonary failure or delayed, progressive cardiac failure, (with or without signs of enteritis).	- After oral ingestion, CCV invade mature epitheliocytes of small intestinal villi and involve entire of small intestine AND INTESTINAL LN. - Virus multiply result in desquamation of mature epithelial cells with villous blunting, loss of absorptive and digestive capacity result in diarrhea.
Clinical Signs	I.P	3-7 days	
	Course	2-12 days	
	Morbidity	High	
	Mortality	High	
	Disease stages/ Forms & Signs	A. Intestinal Form (more 6 w): - Fever, depression, lethargy and anorexia. - Clinical or subclinical infection. - Severe enteritis with vomiting and diarrhea which is often blood tinged due to destruction of epithelial cells of intestinal crypts. - Dehydration, shock and death within 2 days. B. Cardiac Form (less 6 w): - Puppies infected during late gestation or in early neonatal period. - Myocarditis with signs of cardiac arrhythmia, dyspnea, coughing, pulmonary edema - Deaths (20-100%) due to myocardial necrosis and myocardial failure.	Clinical signs Adult - The majority of infections will be inapparent, with no symptoms. - Fever is typically very rare, anorexia and depression. - Sometimes, a single instance of vomiting and a few days of explosive diarrhea (liquid, yellow-green or orange). - Occasionally, an infected dog may also experience some mild respiratory problems. Clinical signs Puppies - Protracted diarrhea and dehydration. - Severe enteritis (inflammation of the small intestine) will occasionally result in death. - Most at risk of developing serious complications with this virus
P/M lesion		- Edema and congestion of abdominal and thoracic lymph nodes; thymic atrophy and bone marrow hypoplasia - A thickened and discoloured intestinal wall; watery, mucoid, or haemorrhagic intestinal contents. - Multifocal necrosis of intestinal crypt epithelium with sloughing. - Pale streaks in the myocardium. - Pulmonary edema, alveolitis, and bacterial colonization of the lungs and liver (complications).	- It is non-specific - Dilated loops of bowel which is filled with gas and watery ingesta - Intestinal mucosa may be congested, or hemorrhagic - Mesenteric lymph nodes is enlarged and edematous.

		3. Canine Parvovirus infection (CPV)	4. Canine coronavirus enteritis (CCVE)
Diagnosis	Field Dx	Depends on case history, clinical signs and P/M lesions.	Depends on case history, clinical signs and P/M lesions.
	Lab. Dx	A. Samples: - Fecal or rectal swabs. - Specimens from internal organs as lung, spleen, thymus or lymph nodes. - Serum and blood. B. Laboratory Examinations: Virus isolation on cell culture Molecular Assays: using nested PCR and real- time PCR, (highly sensitive and specific). Serological Assays: indirect fluorescent antibody test (IFAT), ELISA and SNT (4-fold increase in ab titer 2 weeks apart). ECG: obvious abnormalities. Radiography: showing cardiomegaly. Histopathology: - Destruction of newly formed epithelium resulting in shortening of intestinal villi. - Interstitial fibrosis of myocardium with presence of I/N IB. Electron Microscope or latex agglutination test on feces.	A. Samples: Fecal sample. Intestines and mesenteric l.n. Serum and blood. B. Laboratory Examinations: Virus isolation on cell culture Molecular assays: Using (RT) PCR assay, nested PCR and real-time PCR, (highly sensitive and specific). Serological assays: (IFAT), ELISA and SNT. Histopathology: Desquamation of mature epithelial cells with villous blunting.
Differential Diagnosis (DDx)		With others causes of diarrhea and myocarditis.	The disease is confused with all diseases causes diarrhea
Prognosis		Bad in young puppies	Prognosis is bad in young puppies
Treatment		- No specific treatment but symptomatic and supportive (prevent secondary infection). - Fluid therapy as ringer's 45 ml/kg, B/W, I/V, Glucose 50% in a dose of 0.5 ml/kg. - Broad spectrum antibiotic as ampicillin or gentamicin. - Anthelmintic to fight parasites. - H2 Blockers to reduce nausea. - Non-absorbable oral antibiotic as neomycin to reduce ammonia producing bacteria in intestine. N.B: The survival rate in dogs is about 70 %, but death may result from severe dehydration, a severe secondary bacterial infection, bacterial toxins in the blood, or a severe intestinal heamorrhage.	- No specific treatment but symptomatic and supportive (prevent secondary infection). - Fluid therapy as ringer's 45 ml/kg, B/W, I/V, - Glucose 50% in a dose of 0.5 ml/kg. - Broad spectrum antibiotic as ampicillin or gentamicin. - H2 Blockers to reduce nausea. - Non-absorbable oral antibiotic as neomycin to reduce ammonia producing bacteria in intestine. - Once the dog has recovered, there will usually be no need for further monitoring. But, keep in mind that there may still be remnants of the virus that are being shed in your dog's feces, potentially placing other dogs at risk.
Control		- Segregation of infected dogs and treat them symptomatically and destruction all source of infection. - Cleaning & disinfection with hypochlorite at 1:10 or 1:30.	Segregation of infected dogs and treat them symptomatically and destruction all source of infection. The food and food staff must be clean Cleaning & disinfection of the kennels. Protect your dog from coming into contact with other dog's feces, as much as that is possible.
Vaccination		- Living attenuated vaccines singly or in combination with other canine vaccines. - Three doses at (6, 9, 12 w), giving immunity 1 years and booster annually. - Puppies from non-vaccinated bitch vaccinated for first time at 1-4 w. age and at 6-16 w. age if from vaccinated dam.	Inactivated corona vaccine. - First dose vaccine. Vanguard plus® 5 or 8 with annual repetition. - Puppies from non-vaccinated bitch vaccinated for first time at 1-4 w. age and at 6-16 w. age if from vaccinated dam.

5. Canine Infectious Tracheobronchitis (CITB)		
Synonyms	Kennel cough complex - Enzootic Hepatitis	
Def	- Highly contagious multifactorial disease ch. by acute or chronic inflammation of the trachea and bronchial airways. - Usually a mild, self-limited disease but may progress to fatal bronchopneumonia in puppies or to chronic bronchitis in debilitated adult. - Common seen where dogs are in close contact with each other	
Aetiology	- Multifactorial. - CPI, CAV-2, CD (primary pathogen involved). - CRV1,2,3 & CHV & CAV-1. <i>Pseudomonas</i>, <i>E. coli</i>, and <i>Klebsiella</i> may cause secondary infections (after viral infection). <i>Bordetella bronchiseptica</i> may act as a primary pathogen, especially in dogs < 6 m. - The role of <i>Mycoplasma</i> sp has not been clearly established. - Concurrent infections with several of these agents are common. Predisposing factors Immunosuppression and stress of weaning, extremes of ventilation, temperature, and humidity apparently increase susceptibility to, and severity of, the disease	
Epidemiology	Distribution: Worldwide and present in Egypt. Host rang: (Dogs). Immunocompromised and young ones are more susceptible. Seasonal incidence: Cold seasons.	Transmission: Source: ocular and nasal discharges. Mode: Inhalation & Contact with contaminated objects. Economic and zoonotic impact: Loss of dog's function and deaths of valuable dogs.
Pathogenesis	- Following aerosol exposure virus multiply in epithelium of nasal mucosa, trachea, bronchi, bronchioles & peribranchial lymph nodes. - Initial damage of tracheobronchial mucosa by viral multiplication, this may facilitate colonization of bacteria. - Bacterial-viral synergism or mixed infection makes the situation worse.	
Clinical Signs	- I.P up to 10 days Course (several days - several weeks) Morbidity rate is high Mortality rate is low.	A. Uncomplicated form - Common in adult & dogs remain eating & alert. - Paroxysms of harsh, dry coughing that easily induced by gentle palpation of the larynx or trachea. - Serous to mucopurulent nasal discharge and conjunctivitis. - The dogs may be arch back, open its mouth, retch, and discharge white foamy mucoïd discharge. - Spontaneous recovery within 1-2. w or less or chronic bronchitis. B. Complicated Form - Common in pups or immunocompromised dogs. - More severe signs. - Fever, depression, anorexia. - Purulent nasal discharge. - Productive moist cough indicates a complicating systemic infection & bronchopneumonia. - Death.
PM	- Inflammation of respiratory tract with congestion and consolidation of lungs - Enlargement of bronchial lymph nodes. - The air passages are filled with frothy, serous, or mucopurulent exudate (acute & subacute). - In chronic bronchitis, the air passages contain excessive viscid mucus.	
Diagnosis	Field diagnosis; depends on case history, clinical signs and P/M lesions. Lab. Diagnosis. Sample: - Nasal, nasopharyngeal or laryngeal swabs. - Nasal discharge. - Tracheal washing fluids. - Blood & serum. Laboratory procedures: - Bacterial culture & viral isolation from suspected materials.	Molecular assays: Using (RT) PCR assay, nested PCR and real-time PCR, (highly sensitive and specific). Serological assays: (IFAT), ELISA and SNT. Haematology: neutrophilia, lymphopenia and eosinopenia. Histopathology: - The epithelial linings air passages are roughened and opaque, a result of diffuse fibrosis, edema, and mononuclear cell infiltration. - There is hypertrophy and hyperplasia of the tracheobronchial mucous glands and goblet cells. Radiology in chest: pulmonary hyperinflation, lobar consolidation. Bronchoscopy: inflamed epithelium and often mucopurulent mucus in the bronchi.
DDx	All causes of respiratory distress.	
Treatment	Antimicrobial therapy: Indicated in case of deeper respiratory or systemic bacterial infection (oral or parenteral 10-14 d). Tetracycline 20 mg/kg, b/w, PO every 8.h for 7.d, Trimethoprim-sulfonamide 15 mg/kg, b/w, PO or S/C every 12.h for 7.d, Cephalexin 30 mg/kg, b/w, PO every 12.h for 7.d. Antitussives: Hydrocodone Bitartrates as 0.22 mg/kg, b/w, PO, or Butorphanol tartrate in 0.05-1 mg/kg, b/w, S/C every 6-24. h. It given alone or in combination with bronchodilators (not in complicated form).	Glucocorticoids or prednisolone: To reduce cough and volume of respiratory secretions as 0.25-0.5 mg/kg, b/w every 12.h for 5-7. d. Bronchodilators: Aminophylline dihydrate as 11 mg/kg, b/w, PO every 6-12.h for 5-10. d or Theophylline elixir 5-10 mg/kg, b/w, Po, every 6-12.h, for 5-10. d. Expectorants: Guaifenesin and Volatile Oil are inhaled as vapor to stimulate the secretion of viscous bronchial mucus. Supportive care: Electrolytes and glucose.
Control	- Segregation of infected dogs and treat them symptomatically and destruction all source of infection. - Good practices of cleanliness and sanitation, disinfection of kennel by sodium hypochlorite or quaternary ammonium compounds. - Minimize population density, maximizing ventilation, personnel disinfection.	
Vaccination	Active immunization by vaccines contains <i>Parainfluenza</i> and <i>Bordetella bronchiseptica</i> or Polyvalent one may be used as intranasal or parenteral with annual vaccination. - Puppies from non-vaccinated bitch are vaccinate for first time at 1-4 w. age and at 6-16 w. age from vaccinated dam. - Live attenuated canine distemper virus, live attenuated canine adenovirus 2 and live attenuated parainfluenza virus, live attenuated canine parvovirus1 &2, inactivated <i>Leptospira canicola</i> and inactivated <i>Leptospira icterohaemorrhagiae</i>.	



6. Canine Herpesvirus Infection

Def	- A severe viral infection of canine ch. By: - Puppies: 100 % mortality - Adult: suffer from with upper respiratory infection, ocular disease, vesicular vaginitis or posthitis. - Recovered from clinical disease is associated with lifelong latent infection.	
Aetiology	CHV-1: Family <i>Herpesviridae</i>, subfamily <i>Alphaherpesvirinae</i>, genus <i>Varicellovirus</i>. - Large, enveloped DNA virus. - Sensitive to heat, lipid solvents (such as ether and chloroform) and most disinfectants. - Resist very cold temp. (stable at -70 °C).	
Epidemiology	1- Distribution: Worldwide and present in Egypt. 2- Host rang: (Dogs). Common at 1-3 w and pregnant bitch & rare in older dogs than 4 m. 3- Seasonal incidence: Cold seasons. 4- Economic and zoonotic impact: Loss of dog's function and deaths of valuable dogs..	5- Transmission: a. Source: ocular, nasal discharges and vaginal discharges. b. Mode: - Contact with animal discharges. - Inhalation & ingestion. - In-utero infection from pregnant bitches to fetuses - Venereal.
Pathogenesis	- Infection of susceptible animals' results in replication of CHV in the surface cells of the nasal mucosa, pharynx, and tonsils. - In the case of newborn susceptible pups or other dogs with compromised immune response, viremia and invasion of diverse visceral organs occur. This cause prenatal death, Infertility, still birth or abortion. - Primary systemic infection is associated with a high degree of viral shedding; shedding by latently infected animals after clinical or subclinical recrudescence is of lesser severity and duration. 	
Clinical Signs	I.P: 7 days Course 2 weeks (very short in young puppies and death occurs after ≤ 24 hr). Morbidity rate is high Mortality rate is variable. A. Young Puppies - Deaths usually occur in puppies 1–3 w old, occasionally in puppies up to 1 m old, and rarely in pups as old as 6 months or more. - Fever lethargy, decreased suckling, diarrhea, nasal discharge, conjunctivitis, corneal edema, erythematous rash, rarely oral or genital vesicles. - Viral pneumonia with dyspnea and coughing. - Leukocytosis may be present. B. Older dogs - Mild rhinitis, which may be part of the “kennel cough” syndrome. - Conjunctivitis and corneal ulcers in the absence of other upper respiratory signs. - Vesicular vaginitis or posthitis. - Infected pregnant bitches may abort or deliver a partially stillborn litter; however, they seldom exhibit other clinical signs, and future breeding's are likely to be successful.	
PM	- Focal necrosis and hemorrhages in different organs as lungs, kidney cortex, adrenal glands, liver and intestinal tract. - All lymph nodes are enlarged and hyperemic, and the spleen is swollen. - Marked neutrophilic and mononuclear infiltration is seen in ocular lesions. - Basophilic or acidophilic IN IB are most common in areas of necrosis in the lung, liver, and kidneys.	
Diagnosis	I) Field diagnosis; depends on case history, clinical signs and P/M lesions. II) Lab. Diagnosis; A. Sample: - Respiratory tract and vaginal secretions or swabs. - Specimens from internal organs as kidney, liver, adrenal glands and lymph nodes - Blood & serum..	B. Laboratory procedures: - Viral isolation on cell culture. - Molecular assays: Using PCR assay, nested PCR and real-time PCR, (highly sensitive and specific). - Serological assays: (IFAT), ELISA and SNT. - Histopathology: IN IB - Hematology: Leukocytosis
DDx	- Kennel cough. - ICH. - Canine distemper. - Toxoplasmosis.	
Treatment	- Prognosis is bad in young puppies - No specific treatment but symptomatic and supportive (prevent secondary infection). - Glucose 50% in a dose of 0.5 ml/kg. - Broad spectrum antibiotic as ampicillin or gentamicin. - Antiviral agents such as vidarabine	- Adult dogs with ocular, respiratory, or genital disease often experience mild and self-limiting signs. - Ophthalmic antiviral (drops or ointment) cidofovir (0.5% bid) has been used successfully in primary ocular infection and may be useful for persistent or painful ocular lesions.
Control	- Segregation of infected dogs and treat them symptomatically and destruction all source of infection. - Good practices of cleanliness and sanitation. - Isolation of infected pregnant bitch (3.w prior to parturition) and puppies of up to 3.w age and reared in incubators at 35°C and 50% humidity - Examination of animals before breeding for vesicular vaginitis is advocated.	
Vaccination	No available vaccine.	

7. Infectious Canine Hepatitis (ICH)

Synonyms	Contagious hepatitis or Canine Adenovirus infection	
Def	- Contagious disease of dogs (fatal in puppies) caused by Adenovirus - Vary from a slight fever to severe depression, abdominal pain, vomiting, diarrhea with or without evidence of hemorrhage, corneal opacity known as "blue eye", marked leukopenia, coagulation disorders and death.	
Aetiology	- CAV-1 (antigenically related only to CAV-2, one of the causes of ICT). - Non-enveloped DNA virus - Resistant to lipid solvents and acids & formalin. - It survives outside the host for weeks or months. - Susceptible to 1–3% solution of sodium hypochlorite (household bleach).	
Epidemiology	Distribution: Worldwide and not recorded in Egypt. Host rang: (Dogs, foxes, wolves, coyotes, bears, lynx, and some pinnipeds). - Dogs less than one-year age are more susceptible and more severely affected. Seasonal incidence: There is no seasonal prevalence. Economic and zoonotic impact: Loss of dog's function and deaths of valuable dogs.	Transmission: Source: All body tissues and secretions of dogs during acute stages of the disease as saliva, feces and urine (it may be present in kidney and excreted in urine for 6-9 months post infection) Mode: - Ingestion. - Inhalation (rare). - Contact with fomites including feeding, utensils and hands. - Ectoparasites may contain the virus.
Pathogenesis	- Infection is followed by replication in tonsils and Peyer's patches, other lymphatic tissues. - Blood → Viraemia for 4 – 8 days. - Virus then replicates in vascular cells in many organs, and in hepatocytes, endothelial cells of renal glomeruli (Hepatitis and glomerulo-nephritis) and the cornea and uvea (Corneal and uveal inflammation 'blue eye') and others organs (organ failure and death). Hepatic regeneration: with formation of sufficient serum antibody 7 d. post infection (no virus from blood and liver). Persistently infection: Dogs shed the virus in their urine for up to 6 months with chronic active hepatitis . The flowchart illustrates the pathogenesis and clinical findings of Canine Adenovirus. It starts with 'Virus' at the top, leading to 'Tonsils', 'Cervical lymph nodes', and 'Thoracic duct', all labeled as 'Clinical and lab findings'. From the 'Thoracic duct', the virus enters the 'Blood (viremia)'. This leads to four main pathways: 1. 'Eye': Affects 'Uvea', 'Anterior chamber', and 'Corneal endothelium', leading to 'Immune complexes', 'Severe uveitis', and 'Corneal edema'. This results in 'Sequelae: Glaucoma, Keratoconus, Hydrophthalmos, Phthisis bulbi', and finally 'Ocular recovery' and 'Endothelium heals'. 2. 'Liver': Affects 'Kupffer cells' and 'Hepatocytes'. 'Low titer' leads to 'Centrolobular necrosis', which can result in 'Acute death' or 'Acute hepatitis' (leading to 'Clinical recovery begins'). 'High titer' leads to 'Chronic hepatitis', which can result in 'Mild or inapparent disease' or 'Chronic hepatic fibrosis' (leading to 'Persistent infection'). 3. 'Kidney': Affects 'Glomerular endothelium', leading to 'Immune complex glomerulonephritis', which results in 'Focal interstitial nephritis (viruria)' and 'Persistent infection'. 4. 'Other endothelium': Affects 'Brain, lungs, lymph nodes, etc.', leading to 'Excreted feces', 'Organ failure', 'DIC', and 'Death'. This pathway also leads to 'Virus cleared from tissues, clinical recovery'. A box labeled 'Persistent infection' is shown at the bottom, connected to 'Chronic hepatic fibrosis' and 'Focal interstitial nephritis (viruria)'.	
Clinical Signs	I.P: 4-10 days Course 5-7 days in uncomplicated cases and is long in presence of concurrent infection and in dogs with chronic active hepatitis.. Morbidity: (less than 5%) Mortality: 10%–30% ..	
Clinical Forms	Per acute form: Sudden death due to damage of vital organs as brain and lungs or due to shock or hepatic coma. Acute form: - Biphasic Fever “Saddle type curve“, anorexia, and thirst. - Abdominal pain, vomiting and diarrhea. - Petechiae of the oral mucosa, as well as enlarged tonsils. - S/C edema of the head, neck, and trunk. - Leukopenia. - Hepatic involvement: Abdominal tenderness, distention due to serosanguineous Ascites and Hepatomegaly & Icteric mucous membrane. - Non-suppurative encephalitis (uncommon) due to vascular damage of the brain tissue. - Eye involvement: Corneal edema, ulceration or perforation and anterior uveitis result in blepharospasm, photophobia and serous ocular discharge (Transient uni or bilateral Corneal Opacity or blue eye). - Conjunctivitis, serous discharge from the eyes and nose. - Death due to hepatic insufficiency and hepatoencephalopathy.	
PM	Abdominal cavity contain clear to bright red fluid Petichelial and echymotic hemorrhage on all serosal surface. - The liver is enlarged, dark, mottled in appearance and fibrinous exudate is present on liver surface and interlobar fissures Gall bladder is thickened edematous and has a bluish white opaque appearance. Spleen is enlarged and bulges on the cut surface. Kidney: Focal hemorrhage in renal cortex. Hemorrhage in midbrain and caudal brain stem. Lungs: Multiple gray to red areas of consolidation Eye: corneal opacification. - Dogs surviving acute phase reveal: small, firm and nodular liver (chronic hepatic fibrosis) and kidney have multiple white foci. I/N IB in hepatic in endothelial cells	
Diagnosis	Field diagnosis: depends on case history, clinical signs and P/M lesions.	Laboratory procedures:

	II) Lab. Diagnosis; 1- Sample: 1. Oropharyngeal secretions, swabs from oropharynx-tonsillar crypt. 2. Feces or rectal swabs, urine 3. Liver, spleen, lymph nodes, kidney, brain, eye, bone marrow, CSF. 4. Serum and blood.	1. Virus isolation on cell culture (CPE after 18- 24.h) "Cell clustering & detachment" 2. Molecular assays: using nested PCR and real- time PCR, (highly sensitive and specific). 3. Serological assays: indirect fluorescent antibody test (IFAT), ELISA and SNT. 4. Histopathology: I/N IB in hepatic and endothelial cells 5. Hematology: leukopenia, lymphopenia and neutropenia and later on there are lymphocytosis and neutrophilia. 6. Serum biochemical analysis: increase in activities of ALT, AST and ALP with moderate to marked bilirubinuria, proteinuria, 7. CSF analysis: Increased in protein content. 8. Abdominal paracentesis yields fluid that varies from clear yellow to bright red.
DDx	With canine distemper and others causes of leukopenia.	
Treatment	1. No specific treatment but symptomatic and supportive. 2. Fluid therapy as ringer's 45 ml/kg, B/W, I/V, 3. Broad spectrum antibiotic as ampicillin or gentamicin 4. Glucose 50% in a dose of 0.5 ml/kg, 5. Atropine ophthalmic ointment to decrease ciliary spasm.	6. Decrease of protein intake, rectal enemas 7. Non-absorbable oral antibiotic as neomycin to reduce ammonia producing bacteria in intestine. 8. Oral potassium therapy and ascorbic acid.
Control	Segregation of infected dogs and treat them symptomatically and destruction all source of infection.	
Vaccination	1. Living attenuated or inactivated vaccines singly or in combination with other canine vaccines. 2. Two doses with 3-4 weeks intervals, giving immunity 6 m -1 years. 3. Puppies from non-vaccinated bitch vaccinated for first time at 1-4 w. age and at 6-16 w. age if from vaccinated dam. 4. Live CAV-1 vaccine produces subclinical interstitial nephritis and persistent shedding of vaccinal virus in urine or respiratory signs. 5. Live CAV-2 vaccine, provide cross-protection against CAV-1 with very little tendency to produce corneal opacities or uveitis, and the virus is not shed in urine. 6. Inactivated CAV-1 vaccine doesn't produce any lesions in vaccinated dogs (short immunity). Live attenuated canine distemper virus, live attenuated canine adenovirus 2 and live attenuated Parainfluenzavirus, live attenuated canine parvovirus1&2, inactivated Leptospira canicola and inactivated Leptospira icterohaemorrhagiae.	



BACTERIAL DISEASES OF CANINE

	8. Leptospirosis (Canine typhus, Infectious jaundice.)	9. Canine Brucellosis	
Definition	Contagious, zoonotic bacterial infection of dog, caused by <i>Leptospira interrogans</i> and characterized by acute nephritis and hepatitis, vasculitis.	Contagious, zoonotic bacterial infection of dog, caused by <i>B. canis</i> and characterized by abortion and infertility in females and orchitis, epididymitis and testicular atrophy in male	
Aetiology	Saprophytic: as <i>L. biflexa</i> (live in water & soil and do not infect animal host) Pathogenic: <i>L. interrogans</i> and <i>L. kirschnerido</i> (not replicate outside the host and can survive in water and wet soil.)	<i>Brucella canis</i> mainly - Rarely <i>B. suis</i>, <i>B. melitensis</i> and <i>B. abortus</i>	
Epidemiology	1- Distribution: Egypt 2- Host rang: Dogs of all ages but cats rare to be infected 3- Seasonal incidence: Summer and early fall 4- Transmission: Direct & indirect contact as - Direct contact with urine of infected & carrier animals - Ingestion of contaminated tissue - Placental & venereal transmission 5- Economic impact: Losses from some deaths & zoonotic importance	1- Distribution: Egypt 2- Host rang: Dogs of all ages and sexes 3- Transmission: Direct and indirect contact as: - Ingestion - Inhalation - Mucous mm such as eyes - Vaginal discharges or semen 4- Economic impact: Losses from abortion & zoonotic importance.	
Pathogenesis	Entrance → Leptospira enter the animal body through mm or damaged skin to blood stream → vasculitis then to: kidney causing inflammation, pain, renal failure and inability to produce urine liver causing inflammation and liver disease lung and by toxins lead to Leptospira Pulmonary Haemorrhage syndrome (lung bleeding), 70% mortality.	After Infection → bacteremia then to: Reproductive tissue (Gravid uterus, fetus, epididymis, prostate gland) causing epididymitis, orchitis, infertility and abortion Lymphoreticular Cells (L. Ns, spleen, bone marrow) causing lymphoreticular hyperplasia Intervertebral discs causing Discospondylitis. Eye causing Anterior Uveitis. Kidney causing glomerulonephritis.	
Clinical Signs Disease stages/ Forms & Signs	- Fever, shivering, weakness, depression, ↓ appetite. - Reluctance to move ↑thirst, urination, and inability to urinate - Rapid dehydration - Vomiting with blood - Diarrhea with/without blood in stool - Resp. distress	Male: - Infertility. - Loss of libido. - Epididymitis; Unilateral or bilateral testicular atrophy.	Female Abortion.
		Principle signs - Lethargy. - Generalized lymphadenopathy. - Splenomegaly.	Rare signs - Discospondylitis. - Uveitis. - Meningoencephalitis.
P/M lesion	- Jaundice - Hemorrhage - Blood in intestinal lumen - Infarction in kidney	- Splenomegaly - Hepatomegaly - Enlarged LNs - Scrotal dermatitis and epididymitis	
Diagnosis	I- Field Diagnosis: History, epidemiology, clinical signs and PM II- Lab. Diagnosis Samples: Urine, blood and serum, kidneys & liver from recently dead or slaughtered animal Laboratory procedures: CBC: ↑ WBCs, ↓RBCs and platelets Biochemistry: ↑kidney and/or liver enzymes, ↓ (Na. &Cl) and ↑ (P. & K.) Urine analysis: dilute urine, + ve for protein and inflammation Serology: DNA-PCR and MAT. - DNA-PCR - MAT (Microscopic Agglutination Test). - Detect DNA of Leptospira in urine or blood. - Detect antibodies against <i>Leptospira</i> in bl. - Rapid test and less expensive than MAT; May give false -ve result due to giving antibiotic before the test. - Slower test as need several days before getting the result.	I- Field Diagnosis: History, epidemiology, clinical signs and PM II- Lab. Diagnosis Samples: Placenta, vaginal discharge, fetal organs, blood, serum, semen Laboratory procedures: Isolation: culture on specific media Urine analysis: glomerulonephritis and proteinuria Semen analysis: sperm abnormalities Serology: RSAT, Mercaptoethanol agglutination test, AGID	
Treatment	Specific antibiotic: doxycycline 5 mg/kg, I.V every 12 hrs. for 2 weeks - For dogs that cannot tolerate doxycycline, initial therapy with penicillin is appropriate, but this should be followed by a two-week doxycycline course to eliminate the renal carrier phase of infection - Dogs with severe renal/liver damage require hospitalization for IV fluid therapy	- No treatment effective in elimination of bacteria and once dog infected with <i>B. canis</i> remain infected for life Doxycycline Surgical sterilization of the infected dog will decrease shedding of the organisms into the environment, thereby reducing the risk to other dogs	
Control	- Limiting dog exposure to potential host reservoir sp. or contaminated water source in the area - Fencing and rodent control can limit exposure to wildlife and farm animals - Cleaning and disinfection using iodine-based disinfectant and use of rubber gloves to treat any contaminated material that may be available to avoid human infection - Leptospirosis can be transmitted to people, so owners of dogs that may have the disease should avoid contact between the owner's and their dog's urine, and wear rubber gloves when cleaning up any areas the dog may have soiled. Any areas where the dog has urinated should be disinfected - The four-serovar vaccine provide protection from clinical disease and reduce urine shedding		- First isolation of infected animals and all confirmed infected animals should be eliminated - Testing of new animals before allowing to enter - Hygienic disposal of aborted fetuses and fetal placenta and fluids - Cleaning and disinfection of infected premises and equipment using iodophors and there is no effective available vaccine There is no effective available vaccine

10. Mange in dogs & cats						
Def	Highly infectious and contagious skin disease of all domestic animals, man and birds, caused by different species of mites, characterized by alopecia, scale formation and severe itching dermatitis.					
Aetiology			Epidemiology			
1. <i>Sarcoptic scabiei</i> var <i>canis</i> Sarcoptic mange (canine scabies). 2. <i>Notoedores cati</i>, Notoedric mange (feline scabies). 3. <i>Cheyletiella</i> spp, Cheyletiellosis (walking dandruff). 4. <i>Demodectic cains</i> Canine demodicosis. 5. <i>Demodex cati</i>, Feline demodicosis. 6. White Mite; Otodectic mange (ear canker)			1- Distribution: Worldwide and recorded in Egypt. 2- Host range: wide. 3- Seasonal incidence: There is no seasonal prevalence. 4- Economic and zoonotic impact: Loss of dog's function and deaths of valuable dogs. 5- Transmission: Source: Infected animals or fomites. Mode: Direct; Contact. Indirect contact with fomites.			
Clinical Signs	variable signs according to type of mite					
1. Sarcoptic Mange (canine scabies)	2. Notoedric Mange (feline scabies):	3. Cheyletiellosis (walking dandruff):	4. Canine Demodicosis:	5. Feline Demodicosis:	6. Otodectic Mange (ear canker)	
• Highly contagious disease caused by sarcoptic scabiei var canis (transmit by contact human), • Transmitted by direct contact • The female burrows tunnels in stratum corneum and lay her eggs (life cycle about 17-21. d), • IP is 10.d-8. w,	• Rare highly contagious disease of cats and kitten, caused by <i>Notoedores cati</i> (may infest other animals including man) • Similar to Sarcoptes in life cycle and morphology but is smaller	Highly contagious skin parasite of dogs, cats, humans, and rabbits caused by <i>Cheyletiella</i> spp. mites. Infest dogs & cats	• it is common skin disease, caused by demodectic cains (large number), inhabit hair follicles, sebaceous glands or apocrine sweat glands, • Transmitted from dams to puppies during nursing within the first 72.h after birth, non-contagious.	It is uncommon follicular mite as in canine, caused by <i>Demodex cati</i> .	• It is caused by white mite which can be seen by naked eye. • Infest external ear canal, feet & tip of tail of dogs & cats.	
Lesion						
1. Primary lesions appear as papules which develop thick crusts with secondary bacterial infection and intense or severe pruritus of sudden onset is characteristic due to hypersensitivity to mite products. 2. The lesions start on the ventral abdomen, chest, ears, elbows, and legs and become generalized if no treatment. 3. Chronic cases show severe thickening of the skin with fold formation (wrinkled) and crust build-up, peripheral lymphadenopathy, and emaciation.	There are 1. Crusts. 2. Alopecia. 3. Severe pruritus. 4. Dry grey scales. 5. Skin reddening, thickened, wrinkled & severe excoriation from intense itching. • Lesions present on ears, head & neck and may become generalized	1. Alopecia, itching, sparse fine dandruff in dogs & cats (lumbosacral region or generalized). 2. Dry grey scales, skin reddening, thickened, wrinkled & severe excoriation from intense itching, this lesion is present on ear, back of neck, head, face abdomen and forearm.	1. Localized demodicosis occurs in dogs <1 y old, there are focal area of erythema & alopecia. 2. Generalized demodicosis is a severe disease with generalized alopecia, papules, pustules and crusting (secondary infection of the lesion). 3. Signs of systemic illness as fever, anorexia & lethargy, generalized lymphadenopathy, pyoderma and draining tracts are seen.	1. Localized demodicosis, there're one or several areas of focal alopecia on the head & neck. 2. Generalized disease there are alopecia, crusting and secondary pyoderma of the whole body with other systemic signs.	1. Result in formation of excess brown wax & dry crust in the ear and purulent inflammation in severe cases with head shaking & scratching at ear.	
Diagnosis						
Depend on signs, history, epidemiology, possible exposure and involvement of other animals including man, skin scrapings exam (several scrapings) and fecal flotation may reveal mites or eggs.						
Treatment						
Clipping of the hairs, crusts & dirt's are removed by shampoo or warm water and soap and then an acaricidal dip applied as lime sulphur, highly effective & safe, several dips 5 days apart, amitraz is an effective scabicide, Ivermectin may be used 200 ug/kg, b/w, orally or S/C, two treatments 2.w apart, very effective and usually curative.		Treatment as in sarcoptic mite.		Treatment of local form is by local application of amitraz or rotenone ointment with good prognosis but generalized form is treated by whole-body amitraz dips (0.025 %) every 2.w and clipping the entire hair coat or ivermectin and washing of the animal with shampoo before dipping, secondary infection is treated by specific antibiotic.		Treatment by weekly lime-sulphur dips (2 %) and amitraz
Treatment	The treatment of dogs & cats should be continuing until last egg has hatched and the last mite has been killed. It is important to treat all animals in contact as some of them may be carriers and the environment should be cleaned, becomes dry and spray with parasiticial sprays.			Control	Hygienic management as, building or bedding and other inert materials may be sprayed or left in dry state for 3 w, prophylactic treatment of suspected animals, improve nutritional state, high protein diet and reduce predisposing factors.	

11. Dermatophycosis of Pets		
Def	- Highly infectious contagious fungal disease of superficial dead keratinized tissues of the hair and skin of all animal species and humans, caused by dermatophyte, characterized clinically by alopecia and circular elevated itching plaques on the skin. - It is an infection of keratinized tissue (skin, hair and claw) by dermatophytes	
Aetiology	Dermatophytes, (<i>M. canis</i> , <i>M. gypsum</i> and <i>T. mentagrophytes</i>)	
Epidemiology	1- Distribution: worldwide. 2- Host rang: In young (kitten & puppies) or debilitated one and in longhaired breeds of cat's infection may be persistent and widespread. 3- Seasonal incidence: no specific season for the diseases. 4- Transmission: a. Source: infected animals or fomites. b. Mode: Contact with infected animals or fomites. 5- Economic and zoonotic impact: Zoonotic: disease is transmitted from cats to humans commonly,	
Pathogenesis		
Clinical Signs	Cats	Dogs
	- The disease is seen particularly in young kittens or immunosuppressed & malnourished cats and clinical signs may be mild and difficult to find. - The lesions are present on ear, face & extremities or generalized, typical circular or irregular patches of alopecia (raised edge) with skin thickness & scaling in severe cases, occasionally miliary or cutaneous ulcerated nodules with pruritis may be present. - Cats may show only a few broken hairs or tiny patches on the bridge of nose, lips, chin or ears and the affected areas are covered by whitish scales with excessive desquamation of keratin layer and the lesions are look like cigarette ash in the coat. - Sometimes claws are infected with opaque whitish mottling and rare secondary pyogenic infection to occurs.	- The disease is less common in dogs than in cats - The lesions are in form of alopecic & scaly patches, may be regional or generalized folliculitis with papules & pustules (fine follicular papules on the periphery). - It presents on the head and forequarters as small circular area contain few hairs and crusts and itching may or may not present, these may coalesce forming irregular patchy areas.
PM		
Diagnosis	I) Field diagnosis; depends on case history, clinical signs. II) Lab. Diagnosis. A. Sample: Hair & skin scales from periphery of the lesions in wet preparation using 20 % potassium hydroxide B. Laboratory procedures: - Fungal culture. - Examination with wood's lamp (useful in establishing a tentative diagnosis of <i>M. canis</i>). - Direct microscopical examination.	
DDX		
Treatment	The disease is usually self-limiting, but resolution can be hastened by treatment and to prevent spread of infection to other animals & people. A-Local applications - Clipping the hair, clean and remove or brushing all debris at the lesions and wash with warm water - Application of antimycotic agents as povidone (1:4 in water) or chlorhexidine (0.5-2 %), thibenzole diluted in water 1:28 & lime sulphur 2 % as topical shampoo or rinse, daily every 5-7 days and miconazole or ketoconazole cream, apply twice daily, topical application should be used for at least 2-4. w post clinical cure. B-Systemic treatment: - It applied in chronic or severe cases and in long haired breeds of cats by oral dosing with griseofulvin (expensive, long term and side effect) 15-20 mg/kg, b/w/day in cats for at least 2.w after clinical recovery or until dermatophyte can no longer isolated, ketoconazole is not licensed for use in cats, in dogs used as 10 mg/kg, b/w orally every 24 .h for 3-4. w, it should be taken with fat-containing meal.	
Control	It depends on isolation of infected animals and treated, hygienic measures and vaccination using killed fungal cell wall vaccines for prevention of <i>M. cains</i> .	

12. Dirofilaria Immitis Infection			
Synonyms	Heart worms or filariasis.		
Def	It is infectious parasitic disease of dogs and occasionally cats, caused by Dirofilaria immitis, characterized by respiratory and congestive heart failure signs.		
Aetiology	Dirofilaria immitis is large heart worm, whitish and slender of up to 15 cm long for male and of 30 cm long for female. It is found in right ventricle of heart and pulmonary artery. It is rare to found outside the vascular system as in brain or eye.		
Epidemiology	1- Geographic distribution: The disease is common in tropical and subtropical countries including Egypt. 2- Species affected: Dogs and occasionally cats, fox, tiger and wolf, older dogs are more affected than young one. 3- Seasonal incidence: The incidence increases with seasons of high mosquito population.		
Life cycle, Mode of infection Transmission	1. The adult worm lives in vascular system, the fertilized female sheds unsheathed microfilariae which may circulate in peripheral blood and surviving up to 2.y, 2. In later time microfilariae is concentrated in viscera especially spleen. 3. The microfilariae are sucked up by mosquitoes with blood and migrates to Malpighian tubules and undergoes two moults and then return to salivary system after 13 days post infection and 4. Injected into dogs when mosquitoes take blood meals 5. Theses larvae migrates into tissues and develop to fourth stage larvae and take about 181 days until microfilariae appear again blood stream. 6. Microfilariae may be cross placenta and pups may be born with circulating microfilariae.		
Pathogenesis	Adult worms and microfilariae are live in circulation result in - Mechanical obstruction to circulation. - Impairment function of heart valves. - Hypertrophy and dilatation of right heart damage of blood vessels wall. - Lung emboli and infarcts. - Secondary pneumonia. - Venous congestion of liver and degeneration. - Hepatic failure. - Renal collapse. - Allergic skin conditions.		
Clinical Signs	Cats		Dogs
	- The course of the disease is usually short. - Cough. - Dyspnea. - Chylothorax. - Pulmonary hemorrhage. - Chronic emesis. - Sudden collapse and death.		Mild cases show no harmful effect or no signs. In severe cases Vasculitis which causes formation of blood clots, this result in occlusion and death of lung tissues with appearance of pneumonic signs as slight cough or continuous cough, moist rales on auscultation, heart sound displaces to the right due to enlargement, hemorrhage from respiratory tract which is called "vomiting blood", tendency to tire and loss of condition. Signs of cardiac insufficiency with venous congestion as enlarged liver - Heart attacks where infected dogs collapse suddenly - Peripheral dropsical swellings - Ascites - Vomiting - Jaundice - jugular pulse - pallor mucous membranes - hepatomegaly or splenomegaly on abdominal palpation - vision disturbance - convulsions - dehydration - paralysis - Collapse and death. - Hepatic failure, allergic skin conditions and renal collapse may occur.
PM	Heart worms are found in right ventricle and pulmonary artery, lymph nodes enlargement, lung is mottled with pink, gray and yellow nodules with area of consolidation and edema, congestion and fibrosis of liver and shrunken pale kidney.		
Diagnosis	I) Field diagnosis; Signs, PM, epidemiology and history of the disease. II) Lab. Diagnosis. A. Sample: Blood films, blood and serum. B. Laboratory procedures: - Examination of blood films after staining for microfilariae. - Thoracic radiograph to show right ventricular hypertrophy, dilatation of pulmonary artery. Echo cardiograph to detect worms in right ventricles. - ECG show alteration. - Serological tests as latex agglutination test, ELISA and FAT. III) Differential Diagnosis: Bronchopneumonia, Feline asthma, Lung worms and Cardiomyopathy.		
Treatment	Adulticides as levamisole or sodium thiacetarsamide 2.2 mg/kg, b/w, I/V, every 12.h for four treatments. - To minimized thrombo-embolism we should use Aspirin 5mg/kg, b/w, orally daily for 2-3 weeks before and for several weeks after injection of adulticides agents Prednisolone 1mg/kg, b/w, once daily, it given in concurrent with aspirin, but it is stopped three days before adulticides drugs. Diuretics as furosemide 2mg/kg, b/w, once daily orally and digoxin may be indicated. - After 4.w of adulticides injection we should use Microfilaricides drugs as ivermectin as single dose 50 mcg/kg, b/w or levamisole 11mg/kg, b/w once daily, orally for 10 days. Surgical removal of adult worms from right atrium or vena cava.		
Control	It depends on control of mosquitoes and prophylactic treatment by diethylcarbamazine 5.5 mg/kg, b/w daily orally may be used where mosquitoes are present.		

FELINE INFECTIOUS DISEASES

	13. Feline Panleukopenia (Feline Infectious Enteritis, Cat Fever, and Cat Typhoid)	14. Feline Infectious Anemia (FIA) (Feline eperythrozoonosis haemobartonellosis)
Definition	Is a highly contagious viral disease of domestic and wild Felids caused by the feline parvovirus characterized clinically by: Severe depression, vomiting, diarrhea, hydration, and is often death.	- It is an acute or chronic mild infectious disease of domestic cats which caused by rickettsial agent (<i>Haemobartonella felis</i>) and characterized by fever, splenomegaly, and anemia. - Infected cats are formed antibodies to infected RBCs resulting in autoimmune haemolytic anemia.
Aetiology	The Feline panleukopenia virus (FPV), Feline Parvovirus - A non-enveloped single stranded DNA virus, which is a parvovirus - Is closely related to, the canine parvovirus-2 (CPV-2). - The FP virus is very stable and can remain infectious at room temperature for as long as a year on fomites. - FPV is resistant to many disinfectants. - Inactivation generally requires concentrated hydrogen peroxide solutions.	A Rickettsia (<i>Haemobartonella felis</i>). - It is not usually present in peripheral blood smears. - It appears as coccoid form and occasionally rod or ring form of 1-3 µm in diameter. - It can be seen in RBCs after staining with Giemsa stain. - Sometimes, it may be present free in plasma.
Epidemiology	1-Distribution: The disease is worldwide distributed and reported mainly in Nigeria. 2-Host rang: Panleukopenia is primarily a disease of kittens, but cats of all ages are susceptible. Is most commonly seen in cats 3-5 months of age; death from FP is more common at this age. Raccoons, mink, and foxes, are reservoir for FPV 3-Transmission: A) Sources: The virus is shed from all body secretions during active stages of the disease. Mainly their feces, but may also be present in the vomitus, urine, and saliva. B) Mode of transmission: - Virus is most commonly transmitted by direct contact of susceptible animals with infected cats and their secretions - Mainly fecal–oral route. - Flies and other insect vectors during warm weathers. - Humans can transfer the virus from one cat to another on their shoes, clothing, and hands.	1-Distribution: It is worldwide distributed. It is not reported Egypt. 2-Host rang: All ages and breeds of cats are susceptible, but it is more common in males and the age group of 1-3 years. Factors influencing susceptibility: - Stress factors such as heat stress, concurrent infections, kitting, and metritis increase the susceptibility of cats. - The incidence may increase during summer seasons 3-Transmission: - Blood sucking parasites such as fleas or mosquitoes may transmit the infection mechanically. - Also, transmission through biting of cats and trans-placental infection are suspected
Pathogenesis	- The virus is transmitted via →the fecal–oral route, initially replicates in tissues of the Oropharynx and is then distributed via a cell- free viremia to all tissues. - The virus infects lymphoid tissues → can cause a functional immunosuppression. - Lymphopenia may arise directly as a result of lymphocytosis, but also indirectly following lymphocyte migration into tissues. - FPV destroys the cells of the intestinal crypts causing enteritis, villous atrophy, and malabsorption. - In utero infection of fetus can occur, leading to → fetal death, mummification, abortion, or stillbirth	After infection, the parasite reaches the blood and multiplies inside the RBCs causing destruction of RBCs and consequently anemia.
Clinical Signs Disease stages/ Forms & Signs	I.P: 2-7 days. - Fever (greater than 40°C), which can persist for 24 h or more. - Death occurs in the Peracute form of the disease. - In appetite, a rough coat, and often repeated vomiting. A profuse and persistent diarrhea may develop at approximately the 3rd or 4th day of illness. - Dehydration from severe malabsorption diarrhea. - Abdominal pain, especially when someone touches the abdomen. - An affected kittens are noticeably ataxic, have a wide-based stance and move, in- coordination and tremors, and these signs persist for life. - Infection in pregnant queens can result in fetal mummification, abortion, or stillbirth of neonates.	A) Acute form: - The clinical manifestations include jaundice or anaemic mucous membranes, depression, weakness, dyspnea, lack of appetite, and wasting. - Splenomegaly is prominent sign and spleen can be palpated through the abdominal wall. - Also, paleness of the m. m and tongue is common. B) Chronic form: - Temperature may be normal or subnormal. - There is weakness, depression, emaciation, and dyspnea. Jaundice and splenomegaly are less likely to occur and are not observed in majority of cases. - Feces are yellowish orange. There is no haemoglobinuria. - Protrusion of the third eyelid and abortion may occur.
P/M lesion	- Rough, hair coat and there is evidence of severe dehydration. - The intestine itself is edematous and has petechial or ecchymotic hemorrhages. - The villi are shortened and blunted (villous atrophy) and fibrinous exudates may be seen on surface of the mucosa. - Thymic atrophy is apparent, lymph nodes are pale and edematous, and the bone marrow is gelatinous or liquid in texture - Eosinophilic intranuclear inclusion bodies are formed during the early stages of infection	- Carcasses are pale, dehydrated, and icteric. - There is congestion and enlargement of spleen and liver. - Mesenteric lymph nodes may be enlarged in about half of the cases with hyperplasia of the bone marrow

	13. Feline Panleukopenia (Feline Infectious Enteritis, Cat Fever, and Cat Typhoid)	14. Feline Infectious Anemia (FIA) (Feline eperythrozoonosis haemobartonellosis)
Diagnosis	I- Field Diagnosis • It depends on history, clinical signs of diarrhea with leukopenia. • PM and epidemiology of the disease. II-Lab. Diagnosis A. Samples: • Feces, urine, saliva, blood, and serum. B. Laboratory Examinations: 1. Virus isolation: virus was isolated in secondary Feline Kidney cell cultures 2. Documenting parvovirus antigen in feces by ELISA 3. PCR-based testing of whole blood or feces, facilitating the diagnosis of FPV in those cats that are ELISA negative. 4. Serology to demonstrate Antibodies to FPV ELISA or indirect IFA but not differentiate between infection and vaccination-induced antibodies 5. CBC, revealed, Leukopenia, Lymphopenia, neutropenia, thrombocytopenia.	I- Field Diagnosis This disease can be suspected from the clinical picture such as anemia, post-mortem lesions, epidemiology, and the history. II-Lab. Diagnosis A. Samples: • Blood samples should be collected with and without anticoagulants. • Samples from the bone marrow should be collected. B. Laboratory procedures: 1. Detection of the parasite in blood and bone marrow: Blood smears or bone marrow films can be examined under microscope after staining with Giemsa, 2. Haematological Examinations: There is reduction of haemoglobin concentration (5%), PCV (20%), and RBCs count. 3. Experimental infection: The experimental infection can be performed by inoculation of the suspected blood sample into kittens. The acute signs develop, and the parasite can be detected in the blood samples. 4. Serological Examinations: They have little value.
DDx	The disease is confused with: 1. Salmonellosis. 2. Feline leukaemia virus. 3. Cryptosporidiosis. 4. Toxoplasmosis.	
Prognosis	Bad if the white blood cell count falls below 1000 cells per mL of blood	
Treatment	The goal of supportive treatment is to make the cat feel comfortable and help his or her immune system fight the virus. 1. Symptomatic, anti-emetics, and medication to fight pain. 2. Antibiotics may be given if the cat has developed a bacterial infection 3. Intravenous fluids with electrolytes and nutrients if the cat is dehydrated and needs nourishment. 4. Injections of vitamin B. 5. Plasma or whole blood transfusion.	1. Blood transfusion is the most effective treatment, particularly in the acute anaemic cases (40-65 ml/kg of whole blood), 2. Haematinics containing liver extracts and Vit B complex are helpful. 3. Oxytetracycline (50-100 mg/cat) orally divided into 3 doses daily for 3 weeks is effective. 4. Oral administration of Prednisolone (5 mg/kg) twice daily for 30 days is helpful in decreasing the rate of haemolysis and depression associated with severe anemia.
Control	A) Hygienic measures: - 1. Isolation of suspected Cats FPLV. 2. Cats of unknown status should not be housed together. 3. This non-enveloped virus is very resistant to environmental conditions and many disinfectants, so it need strict hygienic measures. 4. Effective cleaning and disinfection of the environment with concentrated hydrogen peroxide solutions.	The control mainly depends on: • Isolation and treatment of the infected cats, • Detection and treatment of the carrier cats, and insect control.
Vaccination	1. Active immunity is solid and long lasting and can be achieved by both inactivated and modified-live virus (MLV) vaccines. 2. Feline panleukopenia virus antiserum has been used to protect cats before a vaccine-induced, active response is obtained. 3. Modified live vaccines are most commonly used and should be given once at 8-12 weeks of age, and a second time 3-4 weeks later. It has also been advised to give a third dose once the kitten is over 4 months of age, and a first booster 1 year later. 4. Modified-live FPLV vaccines are not recommended in pregnant queens, very young kittens.	

15. Feline Infectious Peritonitis

Definition

It is a progressive highly fatal systemic immune-mediated disease of cats caused by feline coronavirus infection and characterized by inapparent enteric infection with fecal shedding of virus, persistent fever does not respond to treatment, pyogranulomatous tissue reaction, accumulation of inflammatory fluids in body cavities, and high mortality.

Aetiology

- **Feline Coronavirus (FCoV)** is a single-stranded enveloped RNA virus.
- There are two serotypes of **FCoV (I&II)** that differ in cell culture characteristics.
- Coronaviruses are found in many animals and are generally adapted for infecting epithelial cells of the respiratory or gastrointestinal tract.

Epidemiology

- 1- Distribution:** FCoV is universal and distributed worldwide.

 - In many regions, 50% of cats are positive for coronaviral antibodies. However, the majority represent current or past inapparent infection with non-mutated FCoV and only some of these develop into mutated FIP-causing infections. Thus, the prevalence of FIP is much lower.

2- Animal susceptibility:

 - Both domestic and exotic cats are susceptible.
 - The FCoV can infect most wild felids, including the lion, cougar, cheetah, jaguar, leopard, bobcat, sand cat, caracal, serval, and lynx. Cheetahs are especially susceptible to developing FIP.

Factors influencing susceptibility:

 - **Age:** Although cats of any age can be affected, the peak incidence for FIP is between 6 months and 3 years of age. The FCoV infection occurs most often in young kittens (between 6 and 16 weeks of age) after maternal antibodies disperse and the young cats have increased risk for developing FIP. Elder cats may have slightly higher likelihood of developing FIP.
 - Viral load, stress, immune impairment, corticosteroids, surgery, and concurrent disease (e.g., feline leukemia virus or feline immunodeficiency virus) can be risk factors and increase the probability of the virus mutating to a form that can cause FIP.
 - Genetically predisposed cats.

3- Economic Importance: FIP has become the deadliest infectious disease of cats with the decline in prevalence of feline leukemia virus from vaccination.

4- Transmission:

Source of infection:

 - The virus is excreted in feces of cats with inapparent enteric infection or carriers for at least 10 months, and some cats shed persistently for many years, possibly for life.
 - The FCoV can also be excreted in saliva, respiratory secretions, and urine, but these are unlikely to be important sources of infection.
 - Transplacental transmission is possible but uncommon.
 - The environmental contamination with small particles of used litter, contaminated surfaces, food and water dishes, and human clothing, shoes, and hands are important sources of infection.
 - Cats with FIP shed mostly the avirulent non-mutated FCoV (not the virulent mutated one that causes FIP); thus, FIP itself is not directly contagious.

Mode of infection:

 - Transmission most frequently occurs through oronasal contact with virus containing feces or contaminated material from the environment.
 - Contaminated litter and dust particles deposited on the fur are ingested during normal grooming activity.

Pathogenesis

1. The nonmutated enterotropic FCoV is spread via fecal shedding and cats can become infected by ingesting FCoV while grooming.
 2. Once ingested, FCoV enters enterocytes using a spike (S) protein gene on the viral surface and replicates in intestinal epithelial cells.
 3. Most cats can eliminate the virus, and a few may become healthy carriers.
 4. Virus replication can destroy the enterocytes and cause a clinically inapparent infection of intestinal epithelial cells with fecal shedding of virus and may rarely show diarrhea.
 5. Generally, FCoV remains in the enterocytes without causing further illness.
- In cases of FIP**
1. FCoV mutates frequently during replication in the cat's intestinal tract especially in kittens, and this sporadically results in critical genetic mutations that enable FCoV to infect and replicate in macrophages.
 2. The mutated virus strays from the enterocytes, infecting circulating monocytes and tissue macrophages.
 3. Infected macrophages can destroy the virus if they receive the proper immunologic signals. If not, they become incubators.
 4. An immunologically competent host can also attempt to destroy infected macrophages to limit infection.
 5. Macrophages then replicate the mutated coronavirus and carry it to target tissues such as the peritoneum, pleura, kidney, uvea, and nervous system, resulting in widespread immune-mediated vasculitis, and perivascular inflammation, that are the characteristic lesions of FIP.
 6. This followed by circulating immune complexes, complement fixation, cytokine release, apoptosis of activated T lymphocytes, and vascular damage with necrosis and increased permeability.
 7. Eventually, the coagulation system is activated and disseminated intravascular coagulation (DIC) may develop. So, it is not the virus that causes the damage in FIP, but the disease is

Clinical signs

1. The incubation period is variable, ranging from a few weeks to several years. Cats are at greatest risk for developing FIP within the first 6 to 18 months after initial infection with FCoV.
2. The early signs of FIP disease can be quite vague.
3. Despite its name, the lesions of FIP are widespread and not restricted to the peritoneum, cats present initially with nonspecific and non-localizing signs, such as chronic unresponsive fever of unknown origin, anorexia, lethargy, weight loss, vomiting, diarrhea, dehydration, and anemia.
4. As the disease progresses, the signs worsen, and additional clinical signs develop indicating either
 - a. **Body cavity effusions (wet form;** represent approximately **75%**) or
 - b. **Organ-specific abnormalities (non-effusive or" dry"** form; represent approximately **25%** and appears to be more common in older cats).
5. Some cats manifest features of both effusive and non-effusive disease or change over time from one form to the other.

Effusion Signs

- The clinical course is usually acute (days to weeks)
1. **Abdominal Effusion (Peritonitis):** Fluid distension of the abdomen where the effusive FIP involving the peritoneal cavity causes progressive, non-painful, fluid distension of the abdomen. This can be detected by palpation and percussion. Extension of the peritoneal inflammation may involve the gastrointestinal tract (vomiting, diarrhea), hepatobiliary system (jaundice), or pancreas (vomiting due to pancreatitis).
 2. **Thoracic Effusion (Pleuritis):** Dyspnea, tachypnoea, and exercise intolerance are the major presenting signs because lung expansion is restricted by compression from fluid in the pleural space. Cats may prefer a sitting or sternal recumbent posture to facilitate breathing.

Organ-Specific Signs

- The course is often chronic and dangerous (weeks to months).
1. It is characterized by multifocal pyogranulomatous inflammation (inflammatory nodular tissue reaction with the presence of pus) and necrotizing vasculitis in various organs, such as the abdominal viscera (e.g., liver, spleen, kidneys, pancreas, and intestines), eyes, central nervous system (CNS), and lungs.
 2. The degree of organ failure determines the presenting clinical signs.
 3. Abdominal disease
 4. **Kidney Disease:** Pyogranulomatous nephritis causes the kidneys to become enlarged, firm, and irregular ('lumpy').
 5. Icterus and hepatomegaly.
 6. Intestinal pyogranulomatous mass and splenomegaly.

3. Pericardial effusion: Due to fibrinous pericarditis. By auscultation muffled heart sounds are heard. 4. Scrotal swelling: May occur in intact males as a direct extension of the peritoneal inflammation and effusion into the testicular tunics.	7. Pancreatic involvement occasionally cause pancreatitis and, rarely, diabetes mellitus. 8. Mesenteric lymphadenopathy. 9. Omental adhesions and mass 10. Neurologic signs: Multifocal pyogranulomatous meningoencephalitis and myelitis. Ataxia, tremors, vestibular dysfunction, seizures, posterior paresis, hyperesthesia, and behavioral changes are the most common signs. 11. Ocular lesions: Usually bilateral and cause uveitis, corneal edema and sometimes blindness. 12. Pyogranulomatous interstitial pneumonia. 13. Testicular enlargement (orchitis).
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Diagnosis
There is no single reliable confirmatory test for FIP, and the diagnosis is usually suspected from clinical signs and the results of routine laboratory evaluations. A. Isolation and Identification of the virus: By RT-PCR assay of feces or by electron microscopy of feces. However, the PCR assays cannot differentiate between mutated and non-mutated FCoV. B. Histopathology: biopsy of affected tissues is a valuable diagnostic procedure for identifying the distinctive FIP lesions of vasculitis and pyogranulomatous inflammation. Intestinal biopsy shows nonspecific lesions of villous tip injury, with stunting and fusion of villi. C. Serology testing is informative as an epidemiological screening tool and as a diagnostic aid for FCoV and FIP. However, it does not provide a definitive diagnosis of FIP because serology cannot distinguish between the non- mutated and the mutated virus'antibodies. In addition, some cats with FIP may test negative on ELISA. D. Laboratory evaluations (e.g., haematology, serum chemistry, cytology, serology, virology, and Urinalysis) can be useful; however, since they are not specific for the disease, they may provide an indication for a possible diagnosis of FIP. The Laboratory Abnormalities includes: - Anemia (nonregenerative). - Neutrophilic leucocytosis or leukopenia. - Lymphopenia. - Elevated serum protein (hyperglobulinemia). - Elevated serum liver enzymes and bilirubin (also bilirubinuria). - Azotemia of primary renal origin. - Proteinuria of renal origin. - Pyogranulomatous or fibrinous body cavity effusion. - Elevated CSF protein and leukocytes (neutrophils). E. Diagnostic imaging is useful for identifying organ sites of involvement in FIP.

Post-mortem lesions
- The gross findings vary depending on the organs or tissue involved. - The carcass is emaciated with rough coat. - The serosal surface of the abdomen and thorax are covered with diffuse or multifocal necrotic fibrinous exudates. - Pale yellow to dark clear or slightly cloudy thick viscous fluid containing fibrin flakes may be found in the peritoneal and pleural cavities. - Pyogranulomatous lesions (multifocal white to gray plaque-like nodules may be observed in the abdominal and thoracic viscera. - The granulomatous lumps may protrude from the surface of the kidneys. - Cats suffered from neurological signs show lesions in the brain and/or spinal cord.

Treatment
1. No treatment has been proved to reduce the risk of developing FIP in healthy antibody positive FCoV carriers or to lower the high mortality rate or to slow progression of the disease and almost all cats with confirmed clinical FIP eventually die. 2. Various antiviral, immunomodulating, and immunosuppressive drugs have been FIP, but the efficacy is questionable. 3. Anti-inflammatory drugs such as corticosteroids (sometimes used in combination with immunosuppressive drugs), may temporarily suppress symptoms and improve the cat's quality of life. These drugs have no effect on the virus, but have anti-inflammatory and immunosuppressive effects, that control the widespread immune-mediated inflammatory reaction that occurs in FIP. However, they may adversely affect cellular immunity mediated by T lymphocytes and macrophages which may promote the viral infection. Such as Prednisone 2–4 mg/kg, q24h, PO. 4. Platelet aggregation inhibitor: such as Aspirin (10 mg/kg, q72h, PO) that inhibit platelet aggregation caused by vasculitis. 5. Topical corticosteroids and atropine Topical ophthalmic for uveitis such as Prednisone acetate (1%) 2–3 drops/eye q6h and Atropine (1%) 1–3 drops/eye up to q6h for mydriasis. 6. Antibiotics to control complicating bacterial infections. 7. Supportive treatment is needed such as: - Minimizing stress. - Parenteral fluid therapy to maintain hydration. - Nutritional therapy via tube feeding. - Body cavity drainage (thoracentesis) to relieve dyspnea. - Blood transfusion for severe non-regenerative anemia. 8. Euthanasia is appropriate in severely affected cats with poor quality of life.

Prevention and control
- Use good husbandry practices (e.g., good sanitation, ventilation, and feeding practices) and limit feco-oral contamination. - Minimize overcrowding and stress because the high-density housing allows a high level of fecal contamination of the environment, facilitating feco-oral spread of infection. - Control feline leukemia virus with vaccination, testing, and removal. - Do not breed male and female cats that have a history of producing kittens that later developed FIP, because they potentially may inherit a genetic susceptibility to FIP. - Identify and eliminate persistent carriers that continuously shed large amounts of virus by using fecal RT-PCR tests repeated over several months. - Wean the kittens early and remove them from the mother at 5 to 6 weeks of age which coinciding with the time of disintegrating the maternal-derived immunity and isolated from all other cats until at least 16 weeks of age. - Maintain strict quarantine procedures to prevent environmental or fomite transfer of FCoV. Vaccination: because infection may have occurred before vaccination, the success of vaccination is not certain. There is no evidence that would help modify the outcome in cats already positive on antibody testing. It is not advisable to vaccinate a positive cat.

16. Feline Leukemia Virus

Definition

It is a chronic viral neoplastic wasting infectious disease of cats caused by Retrovirus infection that causes inability to develop a normal immune response (known as “**immunodeficiency**”) lymphoma, leukemia, bone marrow suppression, and a variety of other clinical syndromes.

Aetiology

Feline leukemia virus (FeLV) is an RNA retrovirus that is a **transmitted** both **vertically** (mother to fetus) and **horizontally** (cat to cat).

Epidemiology

- 1- **Distribution:** FeLV infects cats worldwide.
- 2- **Animal susceptibility:** Cats are the main host of this disease. Human may be infected with this virus.
- 3- **Factors influencing susceptibility**

a. The FeLV infection varies with age, health, environment, and lifestyle.

b. All ages of cats can be affected. However, Kittens less than 4 months of age are much more susceptible to infection than adults.

c. Resistance develops with age, but even healthy adults can become infected.

d. Newly born kittens act as carriers.

e. All species of cats are susceptible.

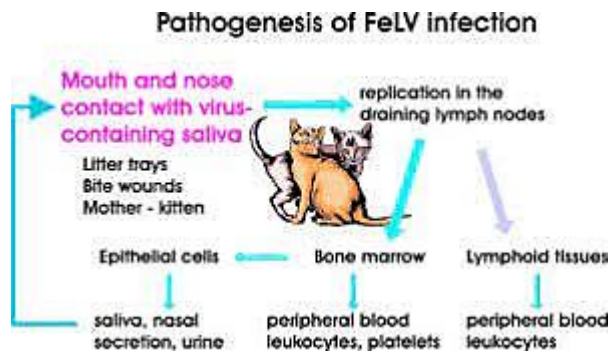
4- **Transmission:**

Source of infection

- a. The FeLV is contagious through prolonged close contact with infected cats.
- b. The virus is excreted in large amount primarily in the **Saliva** of the infected cats and, to a lesser extent, in **Naso-Ocular Secretions** and **Milk**.
- c. Feces, urine, and fleas are less likely sources of infection.

Mode of transmission

- a. Transmission most often occurs through **direct oronasal contact** with infectious saliva through licking during mutual grooming and sharing of food and water bowls or through bite wound inoculation of saliva.
- b. **Transplacental transmission** can occur, but milk-borne infection in nursing kittens is more common; where, kittens can be born infected, or nursing kittens can be infected via milk.
- c. Transmission of FeLV can occur through **contaminated** blood transfusions, needles, surgical & dental instruments.



Pathogenesis

1. After oronasal or percutaneous inoculation, FeLV replicates in local lymphoid tissues and then in systemic lymphoid tissues throughout the body, such as lymph nodes, spleen, and thymus.

2. Viral antigen may become detectable in the blood (**antigenemia**) at this stage.

• **A successful immune response may terminate the infection** at this stage or even before antigenemia is detectable.

3. **If infection progresses**, FeLV infects the bone marrow, leading to circulation of virus infected leukocytes and platelets (**cell-associated viremia**) that are detectable in the blood.

4. Concurrent with the development of viremia, FeLV infects glandular cells throughout the body (e.g., salivary glands, lacrimal glands, mammary glands, and mucosal epithelial glands).

5. This leads to shedding of virus in most body secretions, with especially high concentrations of virus in saliva and the milk of lactating queens.

6. At this stage, FeLV-infected cats are contagious to other cats in close contact, and queens can infect their nursing kittens.
7. **Neutralizing antibodies** are produced and directed against FeLV envelope antigens.

• **Transient infection and recovery occur if the immune response is successful**

8. Bone marrow infection and persistent viremia occur if the virus overpowers the host immune response.

9. **Antitumor response** is mediated by antibody directed against FeLV-associated antigens on the surface of FeLV-induced neoplastic cells.

10. The FeLV-induced neoplasia can be lymphoid or myeloid.

11. **Degenerative and cytopathic effects** on various cells include Bone Marrow Cells (anemia, neutropenia, thrombocytopenia), Lymphocytes (T lymphocyte depletion, lymphoid atrophy, lymphoid hyperplasia), intestinal cells (enteritis), and the Fetus and Placenta (abortion, stillbirth).

12. The immunosuppressive effects of FeLV cause profound immunodeficiency, resulting in susceptibility to a wide variety of opportunistic infections.

13. In addition, FeLV-related immune dysfunction has been associated with immune-mediated and autoimmune diseases.

Sequalae of FeLV exposure

Cats exposed to FeLV can be categorized as:

1- **Noninfected:**

- Many cats exposed to FeLV do not get infected, either because of an inherent or insufficient exposure.

• The most important factor is age-related resistance in cats over 4 m of age.

2- **Persistently infected:**

- Cats with persistent or —progressivell FeLV infection have generalized lymphoid, bone marrow, and polyglandular infection with persistent cellassociated viremia and continuous shedding of virus.

• This infection persists for the remainder of the cat's life with only rare exceptions.

• It leads to fatal FeLV-related disease in most cats; however, kittens less than 4 months of age are most susceptible, and the persistent infection rate can reach 70% in this group.

• Vaccination prevents persistent infection in many cats.

3- **Transiently infected:**

- Cats exposed to FeLV develop a transient —regressivell infection that is subsequently defeated by a vigorous immune response, resulting in a full clinical recovery within 4 to 6 weeks after exposure.

• Transiently infected cats frequently develop latent infection.

4- **Latently infected:**

- Non-viremic cats that —recoverll from transient infection often become latent carriers of non-replicating FeLV for a variable period of time.

• The latent stage of infection is eliminated within 6 to 9 months as part of the normal recovery process, but sometimes take a year or more.

• The significance of FeLV latency is the rare possibility of reactivation either spontaneously or in response to immunosuppression (e.g., corticosteroid-induced). Reactivation of latent-infected queens may transmit FeLV to kittens in utero or during nursing.

Clinical signs

1. The incubation is ranged from one moth to several years.
2. The morbidity rate is low.
3. The mortality rate is up to 100% with long course.
4. The clinical syndromes associated with persistent viraemia can be subdivided into neoplastic and non-neoplastic diseases:

a. **Neoplastic diseases:** FeLV is the cause of the vast majority of lymphomas (15 to 20%), the most common type of tumor occurs in cats.

b. **Non-neoplastic diseases:**

• FeLV infection also has immunosuppressive effects which is secondary to a depletion in lymphocytes or neutrophils, or interference with their function.

• Immunosuppression is the most common manifestation of FeLV infection, with affected cats being more susceptible to coinfection with opportunists or other pathogens, resulting in clinical signs associated with the secondary

Diagnosis

- I- **Field diagnosis:** Feline leukemia can be suspected in the field from anemia, lethargy, anorexia, and enlargement of the lymph nodes, beside the history, PM, and the epidemiology.
- II- **Laboratory diagnosis:**
- A. **Specimens:** The samples include saliva, tears, blood with anticoagulant, organ biopsies or smears, pleural fluids, and serum.
- B. **Laboratory Procedures:**
- Isolation and identification of the virus** by inoculation of the suspected materials into cell cultures. It is the preferred —gold standardII for validating the accuracy of other testing methods. Bone marrow culture can be used to detect latent FeLV infection.
 - The virus antigens particles (FeLV nucleic acid sequences) can be detected in the infected leukocytes, platelets, bone marrow, or biopsy specimens by **PCR** that is considered a highly sensitive confirmatory. PCR may be also useful for detecting latent FeLV infection.
 - ELISA** is considered the preferred screening tests for FeLV because they are rapid, widely available, slightly more sensitive, and detect infection earlier than the IFA test. Using serum or plasma rather than whole blood or saliva and tears are most accurate. However, the IFA test is the preferred confirmatory test for FeLV because it is slightly more specific than ELISA tests and is highly correlated with persistent infection where, the results of virus isolation and the positive IFA results indicate an advanced stage of FeLV infection.
 - Serum Neutralizing Antibody** titers are of limited usefulness in the diagnosis or clinical evaluation of FeLV infection. Where, a positive titer simply indicates prior exposure to FeLV through infection or vaccination, but a titer does not confirm current active infection.
 - Haematological Examinations** that reveal reduction in the reticulocytes count, presence of the nucleated RBCs, presence of immature lymphocytes, and increase the number of WBCs.
 - Experimental infection** by injection of kittens with cell free or cell containing suspension from tumor tissue results in infection.

Post-mortem lesions

- The gross lesions include pleural effusions, and hydrothorax.
- Enlargement in lymph nodes, liver, spleen, and kidneys.
- The wall of the stomach and intestine may be diffusely thickened or contain discrete tumors.

Treatment

- There is **no proven effective treatment for FeLV**, and euthanasia is indicated.
- However, various immune modulator and antiviral drugs have been used.
- Non-specific immune stimulant such as levamisole 2.2 mg/kg orally daily for 3 successive days may be helpful.
- The longevity of FeLV-infected cats is enhanced by general health care, palliative therapy such as antibiotics for secondary bacterial infections that may arise, fluid therapy, and nutritional support, may prolong survival in selected patients.
- Avoid using corticosteroids and other immunosuppressive drugs except when clearly indicated.

Prevention and control

- Improve general wellness care for FeLV-infected cats and take measures to prevent the spread of infection to other cats and to reduce the cat's exposure to infectious agents carried by other animals.
- If the cat lives in a multi-cat household, test all other cats and separate infected and uninfected cats and from kittens, because they are most susceptible.
- Prevent sharing of food and water dishes, litter pans, and any other item that could be a source of cross-contamination.
- The premises and equipment should be regularly disinfectant.
- Vaccination of all uninfected cats by inactivated vaccines or subunit vaccine may be indicated to reduce susceptibility.
- Reduces roaming, aggression, and other stressful behaviors of sexually intact cats.
- Feed cats a nutritionally balanced and complete feline diet. Do not cats feed raw meat, uncooked eggs, or unpasteurized milk, because the risk of foodborne bacterial and parasitic infections is higher in immunocompromised cats.

Pathogenesis of feline leukemia virus (FeLV) showing common entry route of virus, sites of viral replication, and pathological consequences of infection.

